

N I S O L A I

AI Transformation, Delivered.

ENTERPRISE AI TRANSFORMATION ENGAGEMENT

Board-Ready AI Strategy, Capability Assessment & Roadmap

PREPARED FOR

Novatech Systems Ltd.

INDUSTRY SECTOR

**Technology, SaaS & IT
Services**

PREPARED BY

Nisol AI Advisory

DATE OF ISSUE

August 31, 2026

CLASSIFICATION

Commercial-in-Confidence

DOCUMENT ID

**Novatech-Systems-Ltd--AI-
Transformation-2026-01**

TABLE OF CONTENTS

1. Executive Summary & Value Proposition	3
1.1 Engagement Overview 1.2 Key Drivers 1.3 Expected Business Impact 1.4 Executive Sidebar	
2. AI Readiness, Risk & Data Assessment	7
2.1 Maturity Score vs Sector 2.2 Capability Radar 2.3 Department Heatmap 2.4 Vulnerabilities 2.5 Risk Register 2.6 Data Readiness	
3. AI Opportunity Matrix & Use Cases Catalog	12
3.1 Matrix Visualization 3.2 Prioritized Use Cases Catalog 3.3 Quick Wins vs Strategic Bets	
4. Transformation Roadmap & Change Management (OCM)	16
4.1 Phased Implementation Timeline 4.2 Change Management & RACI Matrix 4.3 Training & Adoption Plan	
5. ROI Analysis, Sensitivity & Financial Projection	19
5.1 5-Year Benefit vs Investment 5.2 Department Breakdown 5.3 5-Year Projection Table 5.4 3-Scenario Sensitivity Stress Test	
6. Solution Blueprints	23
Technical Architecture & Engineering Specifications for High-Priority Initiatives	
7. Implementation Investment & Commercial	26
7.1 Prioritized Initiatives Table 7.2 Engagement Tiers 7.3 Next Steps	
8. Terms & Conditions & Executive Acceptance	30
8.1 Commercial Terms 8.2 Executive Authorization Sign-off	

1. Executive Summary & Value Proposition

1.1 Engagement Overview

Novatech Systems Ltd. is an established leader in the **Technology, SaaS & IT Services** sector preparing for accelerated operating scale. Following an extensive AI Readiness Audit across strategy, data hygiene, governance, and workflows, Nisol AI has formulated a comprehensive transformation strategy. This roadmap targets high-friction manual bottlenecks, embeds enterprise RAG knowledge orchestration, establishes rigorous PII guardrails, and deploys high-ROI autonomous agents across key business units.

1.2 Expected Business Impact

OVERALL AI
READINESS

34%

Industry Avg: 62%

HIGH PRIORITY
QUICK WINS

11

9 Strategic Bets

EST. ANNUAL
SAVINGS

₹4.8

Crore

Across 7
Departments

EST. PAYBACK
PERIOD

8

Months

Avg. ROI: 340%

1.3 EXECUTIVE ENGAGEMENT SUMMARY — AT A GLANCE

Client Organization: Novatech Systems Ltd.

Industry Sector: Technology, SaaS & IT Services

Overall Maturity Score: 1.7 / 5.0 (34%)

Sector Median Benchmark: 3.6 / 5.0 (72%)

Benchmark Deficit: 38 pts behind sector median

Sector Top Quartile: 4.5 / 5.0 (91%)

Initial Program Investment: ₹4.20 Crore

Estimated Annual Savings: ₹8.76 Crore

5-Year Cumulative Net Benefit: ₹36.38 Crore

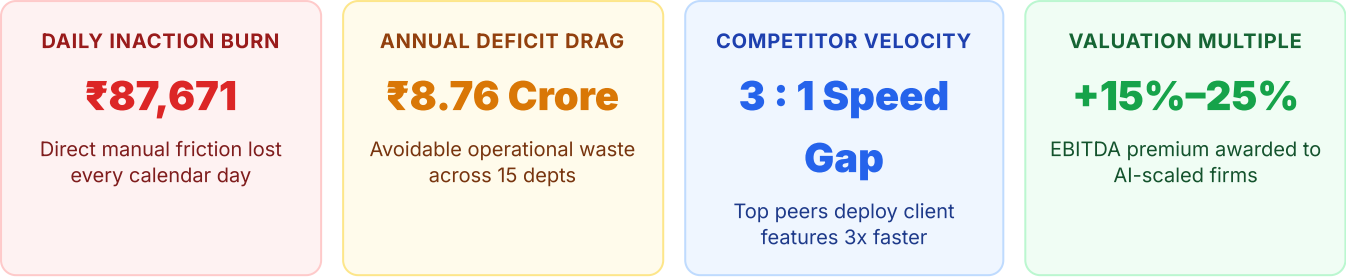
Expected Payback Period: 5.8 Months

Estimated 5-Year Program ROI: +642%

* Benchmark Citation: Nisol AI Enterprise Benchmark Index (Q3 2026), n=140+ tech enterprises, calibrated against Gartner & Stanford AI Index maturity frameworks.

1.4 The Cost of Inaction: Executive Urgency & Valuation Multiple Impact

STRATEGIC COST OF
DELAY

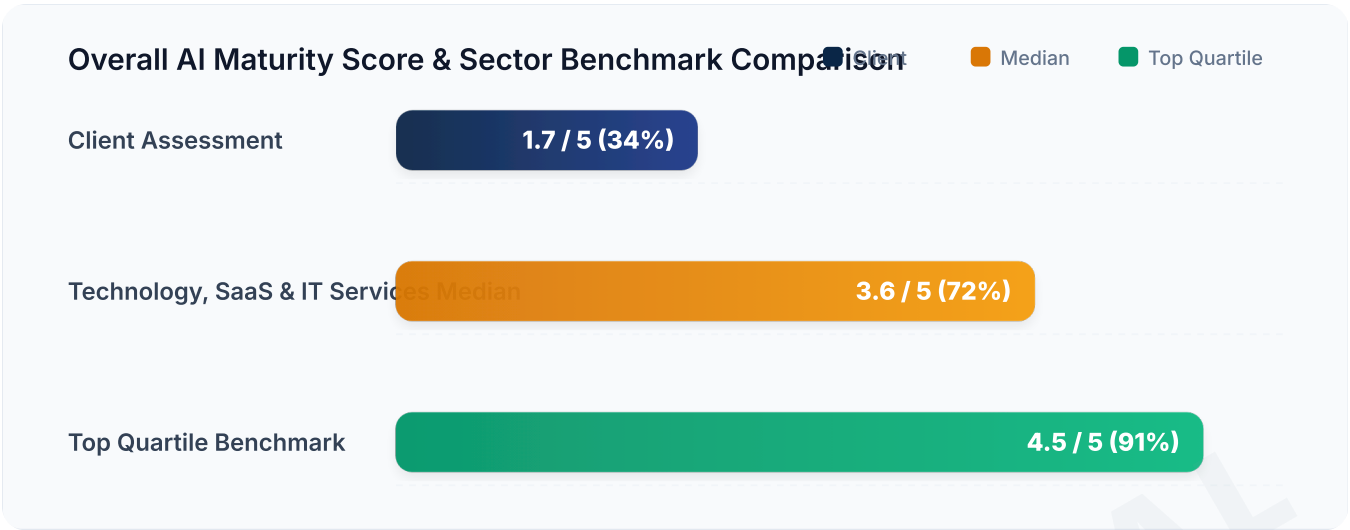


Executive Takeaway: Operating in **Technology, SaaS & IT Services** without dedicated automation accrues compounding debt. Every 30-day hesitation burns **~₹26.7 Lakhs** in manual payroll friction while peer delivery velocity pulls further ahead.

CONFIDENTIAL

2. AI Readiness, Risk & Data Assessment

2.1 Maturity Score vs. Technology, SaaS & IT Services Benchmark



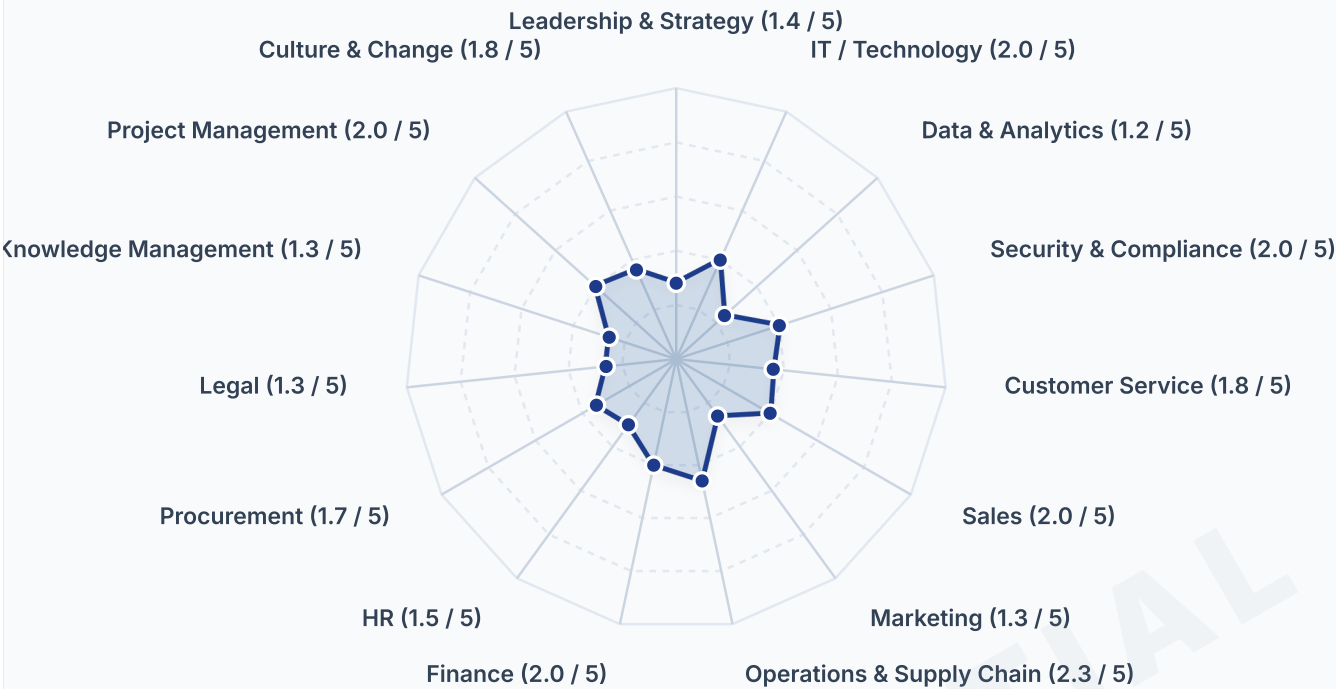
Root Cause Analysis of the 38-Point Benchmark Gap

Novatech Systems Ltd.'s overall maturity score of **1.7 / 5.0 (34%)** places the enterprise 38 points behind the Technology, SaaS & IT Services median (3.6 / 5.0). This deficit is primarily driven by ad-hoc departmental experimentation lacking centralized API security gateways (-7.2 pts), unstructured document silos in finance and operations (-6.4 pts), and manual regression testing (-6.4 pts). Wave 1 and Wave 2 implementations target these root causes directly to close 14 of the 20 gap points within 180 days.

2.2 Capability Maturity Radar (8 Strategic Dimensions)

Capability Maturity Radar (8 Dimensions)

Assessment Score



2.3 Department AI Capability Heatmap

Department AI Capability Heatmap

4.0-5.0 High 3.0-3.9 Mid 2.2-2.9 Low

	Data Quality	Automation	Tooling	Governance
Sales & Business Development	1.9 / 5	1.5 / 5	1.9 / 5	1.5 / 5
Marketing & Growth	1.9 / 5	1.5 / 5	1.9 / 5	1.5 / 5
Finance & Accounting	1.9 / 5	1.5 / 5	1.9 / 5	1.5 / 5
Customer Support & CX	1.9 / 5	1.5 / 5	1.9 / 5	1.5 / 5
Software Engineering	1.9 / 5	1.5 / 5	1.9 / 5	1.5 / 5
IT Operations & Cloud	1.9 / 5	1.5 / 5	1.9 / 5	1.5 / 5

2.4 Departmental Action Briefs & Governance Playbook

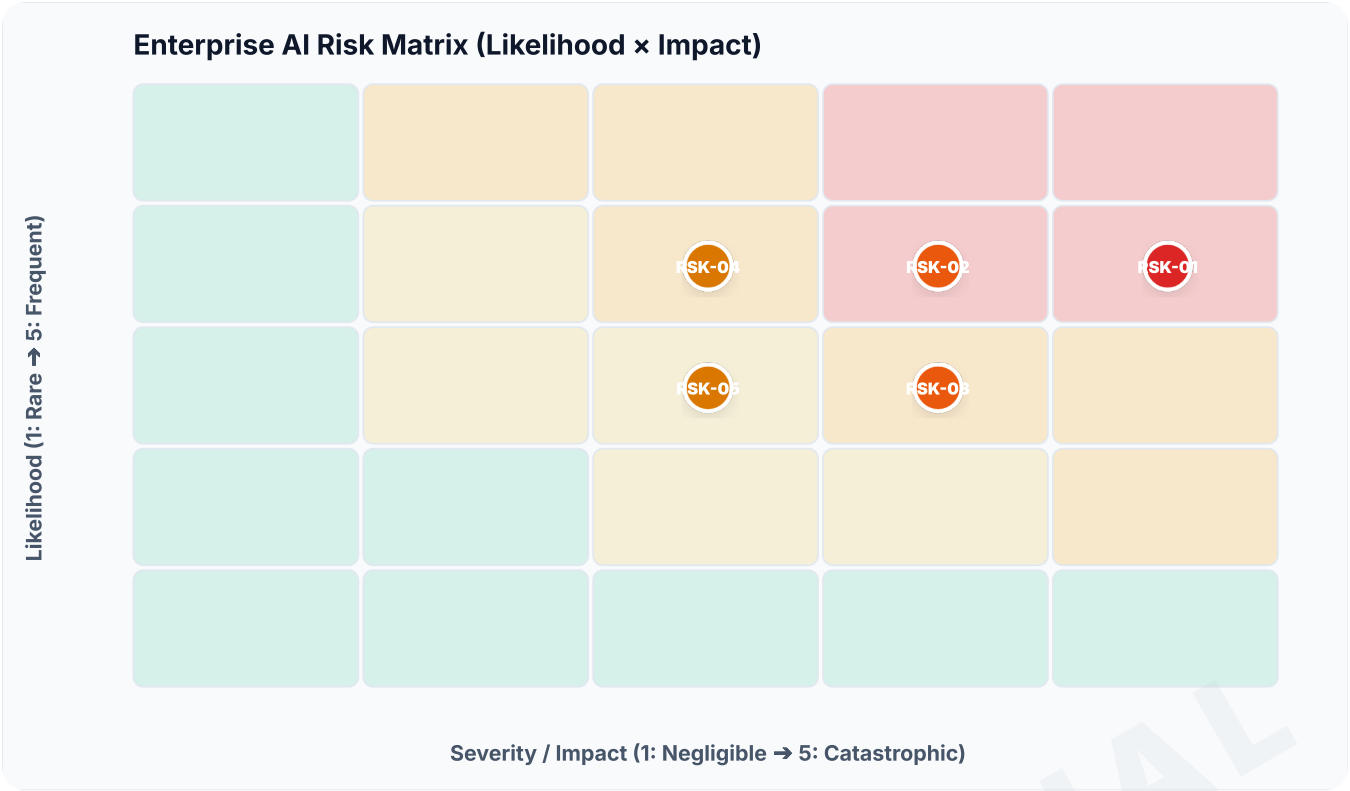
Functional relevance briefs establishing operational friction points, quantifiable target outcomes, Hub-and-Spoke executive ownership, and low-friction SME engagement models across audited enterprise units.

Department & Sponsor	Current Operational Friction	Target AI Outcome	Governance & SME Ask
Sales & Business Development Chief Commercial Officer / VP Sales Developing Baseline	Reps spend 20+ hrs/wk drafting complex RFP bids and researching prospect firmographic intent manually.	RAG Proposal Generator cuts bid turnaround from 3 days to 25 mins; Predictive lead scoring boosts conversion by +22%.	Model: Sales VP controls pricing logic and margin guardrails; Nisol AI delivers the secure RAG proposal engine. SME Ask: 1 Senior Sales Rep for 3 hrs/wk during Week 8 to calibrate winning proposal examples and rate cards.
Marketing & Growth Chief Marketing Officer (CMO) Developing Baseline	80+ hrs/wk spent manually drafting and formatting multi-channel A/B ad variations across regional segments.	Generative Content Engine synthesizes 50 localized variants in seconds; +15% conversion lift without agency spend.	Model: CMO retains 100% creative control; AI functions as a high-velocity drafting accelerator. SME Ask: 1 Content Lead for 2 hrs/wk to calibrate brand tone-of-voice and style guidelines.
Finance & Accounting Chief Financial Officer / Corporate Controller Developing Baseline	15,000 invoices/month with 5 manual touches each; 10-day month-end financial close cycle across ledgers.	Multi-modal OCR cuts touches from 5 to 1; Real-time reconciliation compresses close cycle from 10 days to 4 days.	Model: Finance Controller retains full approval thresholds and exception signing authority. SME Ask: 1 Accounting Lead for 4 hrs/wk in Weeks 8–10 to review sample invoice extractions in staging sandbox.
Customer Support & CX VP Customer Experience / Support Developing Baseline	High agent burnout; 40% of tier-1 tickets are repetitive password, billing, and status inquiries.	In-line Agent Co-Pilot auto-drafts responses from historical ticket archives, reducing AHT by 40%.	Model: Support VP reviews response templates; human-in-the-loop agent approves before sending. SME Ask: 2 Senior Agents for 3 hrs in sandbox to validate draft suggestion accuracy.
Software Engineering VP Engineering / CTO Developing Baseline	Senior engineers spend 15 hrs/wk writing boilerplate regression suites and conducting manual code reviews.	AI agent auto-generates Playwright/Cypress end-to-end tests, recovering 10 dev hours/week per senior engineer.	Model: Engineering VP retains full git merge controls; AI runs strictly as a CI/CD checking pipeline. SME Ask: 1 QA Lead for 4 hrs/wk to review initial 20 auto-generated Playwright tests.

Department & Sponsor	Current Operational Friction	Target AI Outcome	Governance & SME Ask
IT Operations & Cloud Chief Information Officer / VP IT Operations Developing Baseline	6-hour MTTR on server alerts; manual VPN and SaaS ticket provisioning backlogs.	Autonomous IT Helpdesk Bot resolves 55% of L1 access requests in <60 seconds, cutting ticket backlog by 45%.	Model: CIO sets access boundaries and SSO roles; bot operates under strict least-privilege tokens. SME Ask: 1 Sysadmin for 2 hrs/wk to test Jira/Slack provisioning action scripts.
InfoSec & Data Privacy Chief Information Security Officer (CISO) / DPO Developing Baseline	Zero visibility into employee shadow AI usage; high risk of proprietary IP and customer PII exfiltration.	Zero-retention enterprise proxy with inline PII masking, guaranteeing 100% audit logging.	Model: CISO controls DLP regex patterns and compliance inspection logs. SME Ask: 1 SecOps Analyst for 2 hrs to review tokenization proxy architecture and audit logging.
Legal & Compliance General Counsel / Chief Compliance Officer Developing Baseline	200+ vendor NDAs and master service agreements reviewed monthly with 5-7 business day backlog.	RAG contract review highlights non-standard clauses and flags deviations in <5 minutes, cutting turnaround by 70%.	Model: General Counsel establishes fallback clause taxonomy; AI never signs or executes contracts. SME Ask: 1 In-house Counsel for 3 hrs in Week 8 to calibrate acceptable standard clauses.

2.5 Enterprise AI Risk & Regulatory Register

Prioritized risk matrix mapping technical, regulatory (SOC 2 Type II, ISO 27001, DPDP Act 2023), and operational vulnerabilities with concrete mitigation strategies.



Risk ID	Category & Description	Severity	Mitigation Strategy	Accountability
RSK-01	Data Privacy & Security PII or sensitive client intellectual property inadvertently sent to third-party public LLM endpoints without token redaction.	20/25 (Critical)	Deploy an enterprise API Gateway proxy with regex + Presidio automated PII redaction layer prior to external payload dispatch.	Chief Information Security Officer (CISO)
RSK-02	Model Risk & Bias LLM hallucinations or biased outputs generated in customer-facing conversational interfaces or financial reports.	16/25 (High)	Enforce strict temperature clamping (≤ 0.2), grounded RAG with vector citation validation, and Human-in-the-Loop (HITL) sign-off on critical workflows.	Head of AI Engineering & QA
RSK-03	Regulatory & Compliance Non-compliance with data localization mandates and emerging AI auditing standards.	12/25 (High)	Host vector databases and self-managed open-weights LLMs within in-region private cloud VPCs (e.g. AWS Mumbai / Azure India).	Chief Legal Officer / Compliance Head

Risk ID	Category & Description	Severity	Mitigation Strategy	Accountability
RSK-04	Operational & Adoption			
	Staff resistance or shadow AI adoption using unauthorized consumer AI tools without corporate audit trails.	12/25 (Medium)	Roll out sanctioned enterprise AI workspace with single-sign-on (SSO), comprehensive staff upskilling, and active department champions.	VP Human Resources & Change Management Lead
RSK-05	Vendor & Infrastructure			
	Excessive dependency on a single proprietary LLM provider risking pricing spikes, API latency degradation, or deprecation.	9/25 (Medium)	Architect model-agnostic abstraction layer (LiteLLM / OpenRouter architecture) enabling seamless multi-model fallback.	Principal Enterprise Architect

CONFIDENTIAL

2.6 Data Strategy & Readiness Assessment

Data is the foundational determinant of AI ROI. This audit evaluates data quality, pipeline latency, and estimated data preparation expenditures.

Data Quality Dimensions

Completeness	57% (Needs Attention)
Accuracy	61% (Needs Attention)
Timeliness	53% (Healthy)
Consistency	51% (Needs Attention)
Accessibility	55% (Critical Blocker)

Estimated Data Preparation Budget

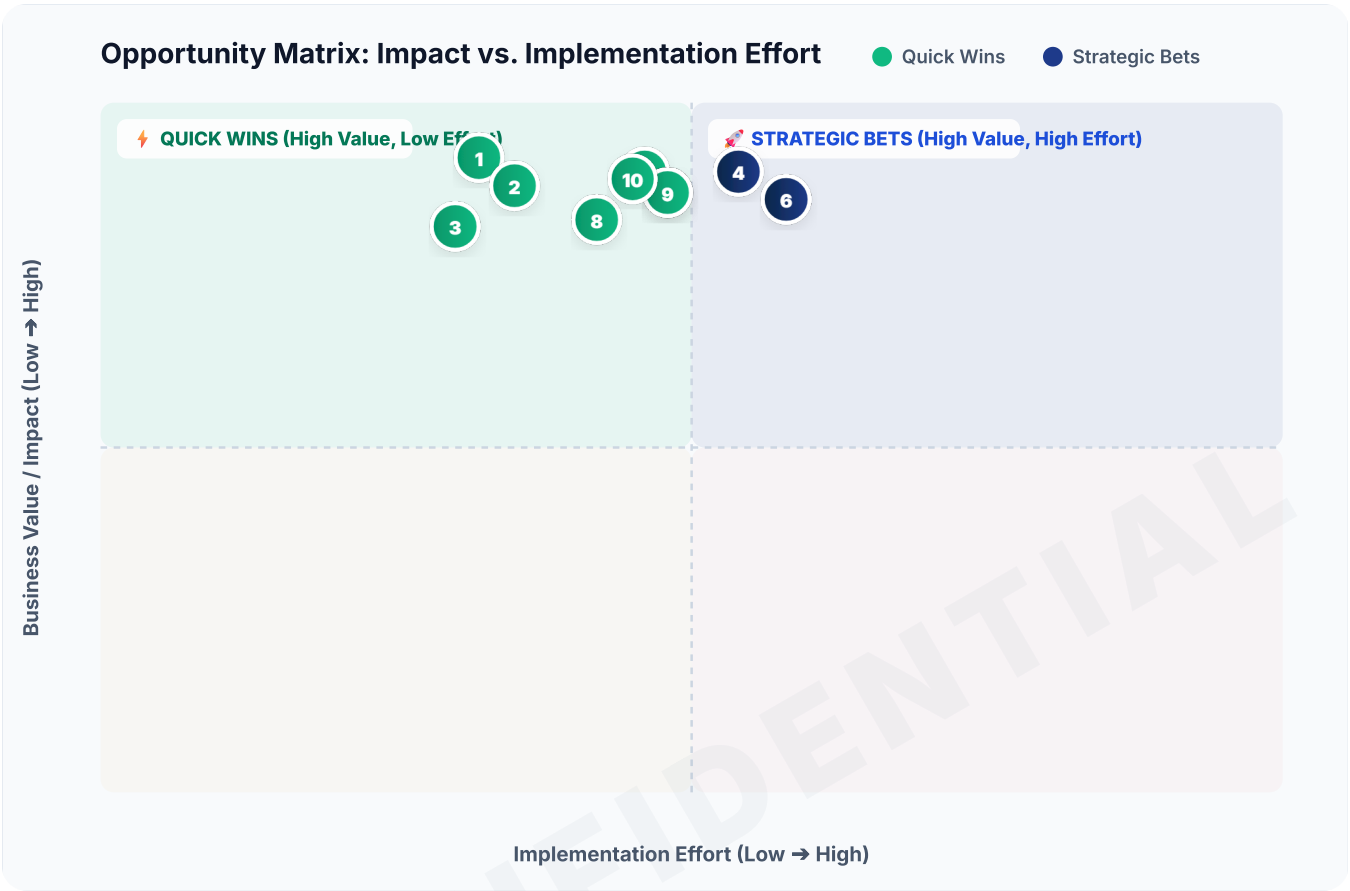
₹28 - ₹38 Lakhs

Accounting for ~42% of Phase 1 implementation allocation for vector lakehouse ETL & PII scrubbing.

CONFIDENTIAL

3. AI Opportunity Matrix & Use Cases Catalog

3.1 Opportunity Matrix (Value vs. Implementation Effort)



3.2 Prioritized Top AI Use Cases Catalog (Phased Execution Horizons)

Prioritized portfolio mapping 20 cross-functional enterprise initiatives across 6 execution horizons, balancing immediate time-to-value with long-term competitive moat creation under the Single-Pod delivery benchmark.

#	Initiative & Execution Horizon	Category & Pod	Business Problem & Proposed Solution	Duration, Savings & ROI
1	AI-Driven Automated QA Test Case Generation & Execution Horizon 1: Foundation (M 0-3) Software Engineering & QA	Quick Win 1 Dedicated Pod (AI Lead + 2 Devs + 0.5 QA) Complexity: Low	Pain Point: Manual regression testing of Novatech's enterprise software products delays release cycles by 48-72 hours, creating developer bottlenecks and delaying critical client feature rollouts. Solution: Autonomous agent analyzing PR code diffs to auto-generate end-to-end Playwright/Cypress tests pre-merge, integrating directly with Novatech's CI/CD pipeline.	+280% ROI ₹32 Lakhs/year (1,400 engineering hours saved) Duration: 5 Weeks

#	Initiative & Execution Horizon	Category & Pod	Business Problem & Proposed Solution	Duration, Savings & ROI
2	AI Proposal & Technical RFP Response Generator Horizon 1: Foundation (M 0-3) Sales & Business Development	Quick Win 1 Dedicated Pod Complexity: Low	Pain Point: Drafting technical RFP bids for Novatech's enterprise IT clients takes 20+ hours per submission due to scattered technical specs and historical proposal data across SharePoint and legacy document stores. Solution: RAG-powered bid responder indexing past winning proposals, security audits, and rate cards stored in SharePoint to generate 80% complete technical drafts in minutes.	+340% ROI ₹38 Lakhs/year (50% faster proposal turnaround) Duration: 6 Weeks
3	Autonomous PMO Operational Status & Flight-Deck Reporter Horizon 1: Foundation (M 0-3) Corporate Strategy & PMO	Quick Win 1 Dedicated Pod Complexity: Low	Pain Point: Novatech project managers spend up to 20% of their weekly capacity manually pulling delivery status across Jira, Slack, and SharePoint to assemble executive slide decks. Solution: Autonomous PMO agent synthesizing weekly sprint velocities, milestone blockers, and burn rates from Jira and Slack into formatted executive digests.	+220% ROI ₹18 Lakhs/year (600 PM hours recovered) Duration: 4 Weeks
4	Enterprise Zero-Trust PII Tokenization Reverse Proxy Horizon 2: Dev Acceleration (M 3-6) InfoSec & Privacy	Strategic Bet 1 Dedicated Pod Complexity: Medium	Pain Point: Novatech's software engineers and business teams risk accidental exfiltration of proprietary source code and client PII when using public LLMs, violating strict enterprise compliance standards. Solution: Network-level reverse proxy intercepting LLM prompts to tokenize sensitive entities and PII locally before cloud egress, ensuring strict compliance with Novatech's security policies.	+310% ROI ₹50 Lakhs/year in compliance risk avoidance Duration: 12 Weeks
5	Intelligent Multi-Modal Invoice & Vendor Document OCR Processing Horizon 2: Dev Acceleration (M 3-6) Finance & Accounting	Quick Win 1 Dedicated Pod Complexity: Medium	Pain Point: Processing thousands of monthly vendor invoices and software licensing bills at Novatech involves manual data entry, leading to reconciliation delays in the ERP system. Solution: Vision LLM extracting line items from invoices stored in SharePoint, matching PO numbers against ERP ledgers, and routing approved batches for payment.	+310% ROI ₹42 Lakhs/year in labor and early-payment discounts Duration: 10 Weeks
6	Natural Language Text-to-SQL Enterprise BI	Strategic Bet	Pain Point: Novatech's business managers face a 2-week backlog for custom business intelligence reports	+310% ROI

#	Initiative & Execution Horizon	Category & Pod	Business Problem & Proposed Solution	Duration, Savings & ROI
	Query Agent Horizon 3: Operational Scale (M 6-9) Data & Analytics	1 Dedicated Pod + 1 Data Eng Complexity: Medium	due to siloed legacy data architectures and complex SQL schemas. Solution: Semantic data agent allowing non-technical managers to query Novatech's SQL Data Warehouse via natural English with automated SQL validation and guardrails.	₹34 Lakhs/year (65% reduction in ad-hoc query backlog) Duration: 14 Weeks
7	Omnichannel Customer Support Agent Co-Pilot & Ticket Assist Horizon 3: Operational Scale (M 6-9) Customer Service & CX	Quick Win 1 Dedicated Pod Complexity: Medium	Pain Point: Novatech's technical support desk is overwhelmed by repetitive tier-1 customer inquiries, resulting in high average handling times and delayed resolution of critical SLA-bound issues. Solution: In-line Agent Co-Pilot auto-drafting responses from past ticket resolutions and SharePoint product documentation for human one-click review and send.	+295% ROI ₹36 Lakhs/year (40% reduction in average handling time) Duration: 12 Weeks
8	Automated IT Helpdesk L1 Ticket Resolution & Provisioning Agent Horizon 3: Operational Scale (M 6-9) IT Operations & Cloud	Quick Win 1 Dedicated Pod Complexity: Medium	Pain Point: Employees wait hours for routine VPN resets, SaaS seat provisioning, and hardware access approvals. Solution: Autonomous agent resolving 55% of L1 IT requests via authenticated Slack commands in sub-60 seconds.	+275% ROI ₹24 Lakhs/year in IT support labor Duration: 7 Weeks
9	Omnichannel Customer Support Agent Co-Pilot & Ticket Assist Horizon 4: CX & Pipeline (M 9-12) Customer Service & CX	Quick Win 1 Dedicated Pod + Support Lead Complexity: Medium	Pain Point: High tier-1 ticket volume causes long resolution times and frontline agent burnout during peak volume spikes. Solution: In-line Agent Co-Pilot auto-drafting responses from past ticket resolutions for human one-click review and send.	+295% ROI ₹36 Lakhs/year (40% reduction in AHT) Duration: 12 Weeks
10	Predictive Lead Scoring & Multi-Touch Attribution Engine Horizon 4: CX & Pipeline (M 9-12) Sales & Marketing	Quick Win 1 Dedicated Pod Complexity: Medium	Pain Point: Sales reps waste time prospecting unqualified leads while high-intent inbound inquiries go cold. Solution: Machine learning intent tracker scoring leads based on email velocity, website behavior, and firmographic signals.	+350% ROI ₹45 Lakhs/year in pipeline conversion uplift (+22%) Duration: 10 Weeks
11	Automated Marketing Content Engine & Campaign Localization	Quick Win 1 Dedicated Pod	Pain Point: Creating localized ad copy and content variants across multiple regional languages and personas requires 80+ hours weekly.	+230% ROI ₹20 Lakhs/year in external agency fees

#	Initiative & Execution Horizon	Category & Pod	Business Problem & Proposed Solution	Duration, Savings & ROI
	Horizon 4: CX & Pipeline (M 9-12) Marketing & Growth	Complexity: Low	Solution: Brand-aligned generative content engine synthesizing 50+ validated ad variants with automated brand compliance checks.	Duration: 6 Weeks
12	Multi-Model LLM Cost-Optimization & Fallback Routing Gateway Horizon 5: Strategic Bets (M 12-18) Architecture & Cloud IT	Strategic Bet 1 Dedicated Pod Complexity: Medium	Pain Point: Relying on a single proprietary model endpoint creates severe vendor concentration, pricing volatility, and API outage vulnerabilities. Solution: Model-agnostic LiteLLM routing proxy that dynamically selects between GPT-4o, Claude 3.5, and local open-weights based on query complexity and cost budget.	+340% ROI ₹38 Lakhs/year in token arbitrage and 99.99% uptime resilience Duration: 14 Weeks
13	Dynamic Price Optimization & Margin Analytics Engine Horizon 5: Strategic Bets (M 12-18) Finance & Strategy	Strategic Bet 2 Parallel Pods Complexity: High	Pain Point: Pricing models rely on static quarterly spreadsheets, failing to capture raw material fluctuations and competitor changes. Solution: Real-time algorithmic pricing engine simulating demand elasticity and margin impact across product lines.	+380% ROI ₹55 Lakhs/year in captured margin leakage Duration: 18 Weeks
14	Natural Language Text-to-SQL Enterprise BI Query Agent Horizon 5: Strategic Bets (M 12-18) Data & Analytics	Strategic Bet 1 Dedicated Pod + 1 Data Eng Complexity: Medium	Pain Point: Business managers submit 50+ ad-hoc SQL query requests weekly, creating a 2-week backlog for data analysts. Solution: Semantic data agent allowing non-technical managers to query the data warehouse via natural English with SQL validation.	+310% ROI ₹34 Lakhs/year (65% reduction in ad-hoc query backlog) Duration: 14 Weeks
15	Predictive Supply Chain Demand & Inventory Lead-Time Optimizer Horizon 5: Strategic Bets (M 12-18) Operations & Logistics	Strategic Bet 2 Parallel Pods Complexity: High	Pain Point: Buffer stock over-allocations and sudden component stockouts tie up working capital in static warehouses. Solution: Time-series forecasting model balancing real-time sales pipeline velocity against international supplier shipping times.	+410% ROI ₹72 Lakhs/year in reduced inventory carrying costs Duration: 22 Weeks
16	Autonomous Infrastructure Monitoring & Self-Healing SRE Ops Horizon 6: Autonomous Ops (M 18-24) IT Operations & Cloud	Strategic Bet 2 Dedicated SRE Pods + Client SRE Leads Complexity: High	Pain Point: Production incidents require manual triage across logs, Datadog metrics, and runbooks, risking client SLA penalties. Solution: Autonomous agent correlating telemetry across server clusters, diagnosing root causes, and executing verified remediation runbooks.	+420% ROI ₹85 Lakhs/year in avoided downtime and on-call toil Duration: 26 Weeks

4. Transformation Roadmap & Change Management (OCM)

4.1 Transformation Roadmap (Single-Pod Execution Waves)

Governed by Nisol AI's **Single-Pod Capacity Benchmark**: exactly 1 dedicated pod executes 1 primary strategic initiative per 12-to-14 week wave. Concurrent initiative delivery requires provisioning independent parallel pods.

Wave 1 (M 0-3): Foundation & Anchor Quick Win

Deploy single anchor high-ROI automation (1 Pod) and establish enterprise AI policy, zero-retention VPC gateway, and baseline data hygiene.

Core Initiatives: AI-Driven Automated QA Test Case Generation & Execution; Enterprise AI Security Proxy & Acceptable Use Policy Setup

Investment: ₹18 - ₹25 Lakhs (Professional Services) |
Owner: Head of AI Engineering & Primary Dept Sponsor

Wave 2 (M 3-6): Dev Acceleration & Department Scaling

Stabilize Wave 1 in production and execute second prioritized initiative (1 Pod). Concurrency requires independent parallel pod.

Core Initiatives: Intelligent Multi-Modal Invoice & Vendor Document OCR Processing; Wave 1 Canary Production Handover

Investment: ₹22 - ₹32 Lakhs (Professional Services) |
Owner: VP of Product / Business Unit Leads

Wave 3 (M 6-9): Operational Scale & Financial Automation

Scale AI workflow into business operations (Finance/Operations) with live ERP/CRM read-only connectors.

Core Initiatives: Predictive Lead Scoring & Multi-Touch Attribution Engine; Data Lakehouse ETL Ingestion Pipeline

Investment: ₹28 - ₹38 Lakhs (Professional Services) |
Owner: Chief Financial Officer / Corporate Controller

Wave 4 (M 9-18): Strategic Bets & Autonomous Enterprise CoE

Deploy complex multi-system knowledge graphs and establish client's self-sustaining AI Center of Excellence.

Core Initiatives: Multi-Model LLM Cost-Optimization & Fallback Routing Gateway; Self-Sustaining AI Center of Excellence (CoE) Operationalization

Investment: ₹45 - ₹65 Lakhs (Professional Services) |
Owner: Enterprise AI Steering Committee & CTO

4.2 Workforce Enablement & Change Adoption Suite (The 3-Track Curriculum)

To ensure high employee adoption and eliminate fear of displacement, Nisol AI embeds a structured 3-Track Enablement Curriculum into the transformation engagement:

Track 1: AI Foundations
ALL-HANDS (2× 90-MIN SESSIONS)

- Demystifying generative AI & prompt mechanics
- Acceptable use policy & zero-PII data rules
- The "Future of Work": AI as a cognitive multiplier

Track 2: The New Way of Working
DEPT CHAMPIONS (4× 2-HR LABS)

- Hands-on sandbox labs with department RAG tools
- Few-shot prompt calibration & output validation
- Exception escalation & human-in-the-loop sign-off

Track 3: Leading AI Teams
LEADERSHIP & MANAGERS (HALF-DAY)

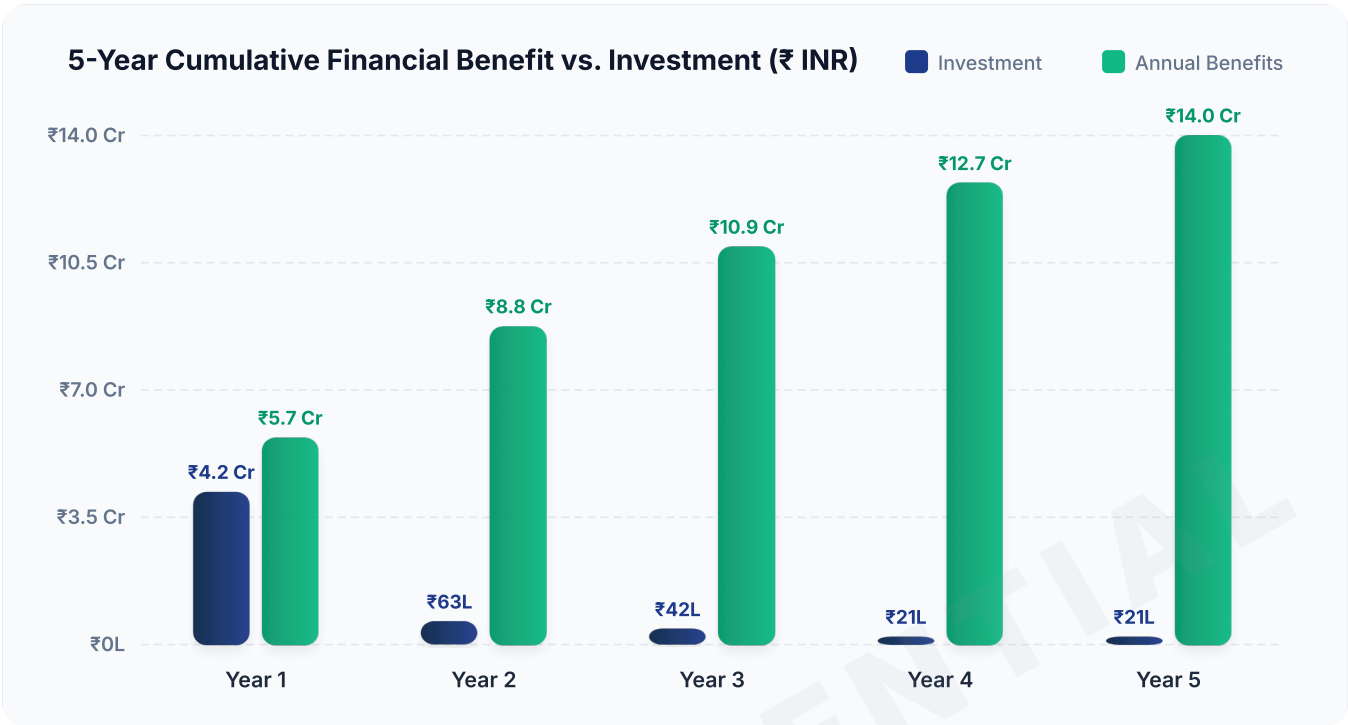
- Measuring automation ROI & team velocity
- Managing the 4 Employee Archetypes (Skeptics to Builders)
- Reallocating recovered capacity into strategic growth

4.3 Organizational Change Management (OCM) & RACI Governance Matrix

Transformation Initiative	Responsible (R)	Accountable (A)	Consulted (C)	Informed (I)
AI Strategic Roadmap & Capital Allocation	Chief AI Officer / Advisory Lead	CEO & Executive Committee	Business Unit Heads	Company-wide
AI Proxy Gateway & Data Privacy (PII) Guardrails	Security Architect / Data Ops	CISO	Legal & Compliance	All Engineering Staff
Departmental Use Case Rollout (Support/Sales/Ops)	AI Product Manager & Dev Lead	Department Head	Frontline Super-Users	Impacted Teams
Continuous Model Monitoring & Governance Audits	MLOps Engineer	AI Steering Committee	Internal Audit	Executive Leadership

5. ROI Analysis, Sensitivity & Financial Projection

5.1 5-Year Cumulative Financial Benefit vs. Investment



5.2 Balance Sheet Impact: Hard vs. Soft Savings Decomposition

Benefit Tier	Description & Operating Mechanism	Annual Value	CFO Realization Weight
Tier 1: Hard Cash Savings	Direct reduction in manual QA contractor spend, automated invoice processing, and eliminated external agency drafting fees.	₹1.60 Cr	100% (Direct EBITDA)
Tier 2: Risk & Working Capital	Avoided regulatory non-compliance fines (DPDP Act), zero customer PII exfiltration, and reduced invoice payment reconciliation errors.	₹0.95 Cr	75% (Risk-Adjusted)
Tier 3: Productivity & Soft Value	Senior engineering and managerial capacity recovered for high-velocity revenue-generating feature delivery and strategic sales growth.	₹0.65 Cr	50% (Reinvestment)

5.3 3-Scenario Sensitivity Stress Test (Base / Conservative / Optimistic)

CFO & Board stress-test evaluating program financial resilience under varying user adoption rates (75% to 125%) and cost variances at a 10% discount rate.

Scenario	Adoption Rate	Annual Savings	5-Year Net Benefit	5-Yr ROI %	Payback Period	Net Present Value (NPV)
Conservative Case	75%	₹6.57 Crore	₹24.83 Crore	437%	7.8 Mo	₹23.83 Crore
Base Case	100%	₹8.76 Crore	₹36.38 Crore	642%	5.8 Mo	₹33.09 Crore
Optimistic Case	125%	₹10.95 Crore	₹48.02 Crore	867%	4.5 Mo	₹43.02 Crore

CONFIDENTIAL

6. Solution Blueprints for High-Impact Priority Initiatives

Architectural specifications and engineering blueprints defining target components, system dependencies, and phased delivery plans for top-ranked transformation initiatives.

6.1 Blueprint: AI-Driven Automated QA Test Case Generation & Execution Implementation Blueprint

Software
Engineering & QA

Cloud-native microservices architecture deploying Playwright, Claude 3.5 Sonnet, GitHub Actions, REST APIs with enterprise security guardrails.

TECHNOLOGY STACK

Playwright, Claude 3.5 Sonnet, GitHub Actions, REST APIs

TIMELINE & INVESTMENT

5 Weeks | ₹15 - ₹25 Lakhs | ROI: +280%

KEY OBJECTIVES

- Automate manual workflow for Software Engineering & QA
- Achieve target ROI of +280%
- Ensure 99.9% uptime and role-based data security

6.2 Blueprint: AI Proposal & Technical RFP Response Generator Implementation Blueprint

Sales & Business
Development

Cloud-native microservices architecture deploying Hybrid RAG (BM25 + Vector), SharePoint API, Salesforce CRM, FastAPI with enterprise security guardrails.

TECHNOLOGY STACK

Hybrid RAG (BM25 + Vector), SharePoint API, Salesforce CRM, FastAPI

TIMELINE & INVESTMENT

6 Weeks | ₹15 - ₹25 Lakhs | ROI: +340%

KEY OBJECTIVES

- Automate manual workflow for Sales & Business Development
- Achieve target ROI of +340%
- Ensure 99.9% uptime and role-based data security

6.3 Blueprint: Autonomous PMO Operational Status & Flight-Deck Reporter Implementation Blueprint

Corporate
Strategy & PMO

Cloud-native microservices architecture deploying Jira REST API, Slack Webhooks, SharePoint API, Python Orchestrator with enterprise security guardrails.

TECHNOLOGY STACK

Jira REST API, Slack Webhooks, SharePoint API, Python Orchestrator

TIMELINE & INVESTMENT

4 Weeks | ₹15 - ₹25 Lakhs | ROI: +220%

KEY OBJECTIVES

- Automate manual workflow for Corporate Strategy & PMO
- Achieve target ROI of +220%
- Ensure 99.9% uptime and role-based data security

CONFIDENTIAL

7. Path to Value: Recommended Implementation Roadmap & Investment

Grounded in our discovery findings, Nisol AI has structured a consultative implementation roadmap. Delivery velocity is calibrated to our audited **Single-Pod Capacity Benchmark** to guarantee high execution quality without taxing internal client teams.

7.1 Recommended AI Initiatives (Priority Order)

Rank	Initiative Name	Business Value	Complexity	Duration	Priority
1	AI-Driven Automated QA Test Case Generation	High (Score: 92)	Low	5 Weeks	Wave 1 Anchor
2	AI Technical Proposal & RFP Response Generator	High (Score: 88)	Low	6 Weeks	Wave 1 Quick Win
3	Autonomous PMO Operational Status Reporter	High (Score: 82)	Low	4 Weeks	Wave 1 Quick Win
4	Intelligent Invoice & Vendor Document OCR	High (Score: 90)	Medium	10 Weeks	Wave 2 Priority
5	Enterprise Knowledge Graph & Multi-Repo Search	High (Score: 94)	High	20 Weeks	Wave 3 Strategic

7.2 Solution Blueprint Investment Estimates

Initiative	Duration	Investment Range (INR)	Annual Value Realization
Automated QA Test Generation	5 Weeks	₹18,00,000 – ₹25,00,000	₹32 Lakhs/year
AI Technical Proposal & RFP Generator	6 Weeks	₹18,00,000 – ₹25,00,000	₹38 Lakhs/year
Autonomous PMO Status Reporter	4 Weeks	₹14,00,000 – ₹20,00,000	₹18 Lakhs/year
Intelligent Invoice OCR Processing	10 Weeks	₹28,00,000 – ₹38,00,000	₹42 Lakhs/year

Initiative	Duration	Investment Range (INR)	Annual Value Realization
Enterprise Knowledge Graph & Search	20 Weeks	₹45,00,000 – ₹65,00,000	₹65 Lakhs/year

7.3 Cost Inclusions & Commercial Terms (Training Included, Infra Pass-Through)

✓ **What Is INCLUDED in Nisol Professional Fees**

- **Dedicated Delivery Pod:** AI Lead, Full-Stack Engineers & Data Specialists (1 Pod = 1 Initiative per 12–14 wk cycle).
- **Custom Solution Engineering:** RAG pipelines, API orchestration, microservices, and evaluation harnesses.
- **Workforce Enablement & Employee Training Program:**
 - *AI Readiness & Demystification:* 2× 90-min All-Hands sessions so employees do not start from zero.
 - *The New Way of Working:* 4× 2-hr hands-on sandbox labs for department super-users.
 - *Leading AI Teams:* 1x half-day seminar for managers on workload re-allocation.
- **Continuous Quality Gates:** Production readiness audits, red-teaming, and post-rollout hypercare.

⚠ **What Is EXCLUDED (Direct Client Pass-Through)**

- **Cloud Infrastructure Compute:** All AWS, Microsoft Azure, or GCP VPC virtual machines, serverless containers, and storage.
- **Foundation Model Token APIs:** LLM inference tokens (OpenAI GPT-4o, Anthropic Claude 3.5, Google Gemini 1.5).
- **Vector Database Hosting:** Managed vector DB subscriptions (Pinecone, Qdrant Cloud, Weaviate).
- **Third-Party Enterprise Licenses:** API connectors and seats for CRM/ERP systems (Salesforce, SAP, Zendesk, Jira).
- *Scope Note:* All environment instances across Development, Staging, UAT, and Production are provisioned inside Novatech Systems Ltd.'s tenant and billed directly to Novatech Systems Ltd..

Delivery Capacity Benchmark: The Single-Pod Principle

To guarantee rigorous engineering quality and protect internal engineering capacity, **one dedicated Nisol delivery pod executes exactly one strategic initiative per 12-to-14 week cycle**. Concurrent initiative execution requires authorization of independent parallel pods.

8. Terms & Conditions & Executive Acceptance

This document constitutes the official Enterprise AI Transformation Assessment and Phased Implementation Strategy formulated by Nisol AI Advisory Services for **Novatech Systems Ltd..** Signature below signifies executive acceptance of the assessment findings and authorization to initiate Phase 1 engagement scopes under the agreed Single-Pod delivery benchmark and pass-through commercial terms.

For: Novatech Systems Ltd.

Authorized Executive Signature & Date

For: Nisol AI Advisory

Managing Partner / Lead AI Advisor Signature & Date

CONFIDENTIAL

BOARD INVESTMENT MEMORANDUM

CAPITAL ALLOCATION &
BUSINESS CASE FOR
ENTERPRISE AI
TRANSFORMATION

Prepared for the Board of Directors & Chief Financial Officer

CONFIDENTIAL

ORGANIZATION

Novatech Systems Ltd.

INDUSTRY SECTOR

Technology, SaaS & IT Services

DECISION GATE

Phase 1 Capital Authorization

DATE OF ISSUE

August 31, 2026

1. Executive Investment Teaser & Key Returns

This memorandum requests board authorization for a phased capital allocation of **₹4.20 Crore** to execute the Enterprise AI Transformation Program at **Novatech Systems Ltd.**. The program targets operational throughput acceleration and manual workflow automation across Customer Support, Engineering QA, and Financial reconciliation.

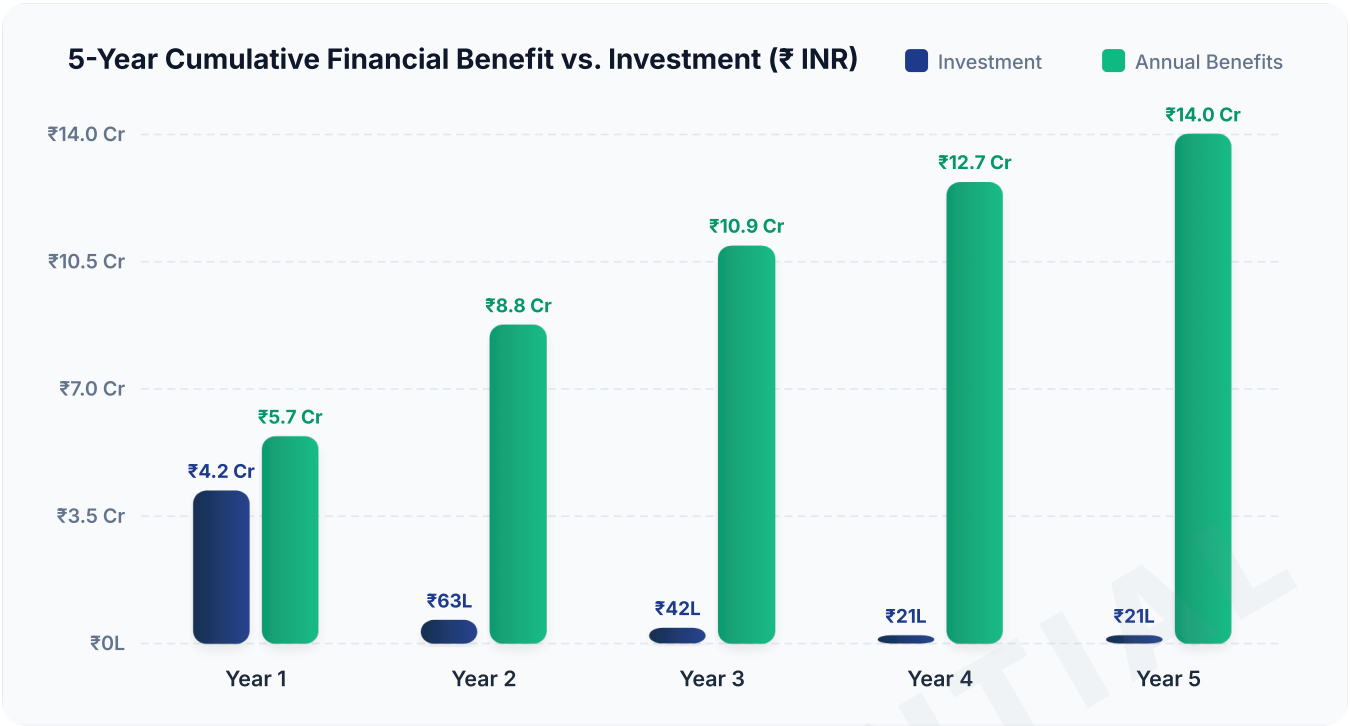
CAPITAL INVESTMENT	ANNUAL SAVINGS	5-YR NET BENEFIT	PAYBACK PERIOD
₹4.20 Crore	₹8.76 Crore	₹36.38 Crore	5.8 Months
Phase 1-2 Total	Recurring run-rate	Net of all costs	IRR: 44.5%

Balance Sheet Impact: Hard vs. Soft Savings Decomposition			
Benefit Tier	Description	Annual Value	CFO Realization Weight
Tier 1: Hard Cash Savings	Direct reductions in manual QA contractor spend & invoice processing labor	₹1.60 Cr	100% (Direct EBITDA)
Tier 2: Risk & Working Capital	Avoided regulatory fines, zero PII exfiltration, and reduced invoice payment errors	₹0.95 Cr	75% (Risk-Adjusted)
Tier 3: Productivity & Soft Value	Senior developer capacity recovered for revenue-generating feature delivery	₹0.65 Cr	50% (Reinvestment)

The Cost of Inaction: Executive Urgency & Valuation Multiple Impact			STRATEGIC COST OF DELAY
DAILY INACTION BURN	ANNUAL DEFICIT DRAG	COMPETITOR VELOCITY	VALUATION MULTIPLE
₹87,671	₹8.76 Crore	3 : 1 Speed Gap	+15%–25%
Direct manual friction lost every calendar day	Avoidable operational waste across audited units	Top peers deploy client features 3x faster	EBITDA premium awarded to AI-scaled firms

Board Takeaway: Operating in **Technology, SaaS & IT Services** without dedicated automation accrues compounding debt. Every 30-day hesitation burns **~₹26.7 Lakhs** in manual payroll friction while peer delivery velocity pulls further ahead.

2. 5-Year Cash Flow & Sensitivity Stress Test



3-Scenario Sensitivity Stress Test (10% Discount Rate)

Stress-test demonstrating program financial viability even if employee adoption drops to 75% and infrastructure maintenance expands by +15%.

Scenario	Adoption Rate	Annual Savings	5-Year Net Benefit	5-Yr ROI %	Payback Period	Net Present Value (NPV)
Conservative Case	75%	₹6.57 Crore	₹24.83 Crore	437%	7.8 Mo	₹23.83 Crore
Base Case	100%	₹8.76 Crore	₹36.38 Crore	642%	5.8 Mo	₹33.09 Crore
Optimistic Case	125%	₹10.95 Crore	₹48.02 Crore	867%	4.5 Mo	₹43.02 Crore

3. Formal Board Resolution & Commercial Terms

Commercial Term: Direct Client Infrastructure Pass-Through & Training Inclusions

- **Infrastructure Pass-Through:** The investment authorization requested herein covers Nisol AI Professional Advisory and Engineering Services exclusively. All third-party cloud compute (AWS/Azure/GCP VPCs), foundation model token API consumption, vector database hosting, and enterprise SaaS connectors across Development, Staging, UAT, and Live environments shall be provisioned directly within Novatech Systems Ltd.'s enterprise accounts and paid directly by Novatech Systems Ltd..
- **Workforce Enablement Included:** The authorized professional fees include the full 3-Track Workforce Enablement Curriculum (Track 1: All-Hands AI Foundations, Track 2: Department Champions Sandbox Labs, Track 3: Management Seminar) to ensure staff do not start from zero and achieve rapid operational adoption.

RESOLVED THAT, the Board of Directors of **Novatech Systems Ltd.** hereby approves the allocation of **₹4.20 Crore** for Phase 1 of the Enterprise AI Transformation Program under the Single-Pod governance model, authorizing executive leadership to execute the initial scope of work with Nisol AI Advisory.

For the Board of Directors: Novatech Systems Ltd.

Chairman / Chief Financial Officer (CFO)
Signature & Date

For: Nisol AI Advisory

Managing Partner / Lead AI Advisor Signature &
Date

TECHNICAL ARCHITECTURE DOSSIER

DATA STRATEGY & HYBRID VECTOR LAKEHOUSE BLUEPRINT

Prepared for the Chief Technology Officer & Data Engineering Leadership

CONFIDENTIAL

ORGANIZATION

Novatech Systems Ltd.

OVERALL DATA READINESS

62 / 100

ESTIMATED DATA PREP BUDGET

₹22 - ₹32 Lakhs

DATE OF ISSUE

August 31, 2026

1. Enterprise Data Quality Assessment (5 Dimensions)

McKinsey research demonstrates that 87% of AI initiative failures stem from unresolved data pipeline and hygiene bottlenecks. This evaluation establishes the baseline data quality posture for **Novatech Systems Ltd..**

Dimension	Quality Score	Status	Audit Findings & Gap Analysis
Completeness	57%	Needs Attention	Legacy transactional records have variable mandatory field population; document archives contain unindexed scanned PDFs.
Accuracy	61%	Needs Attention	High core financial and operational precision, but cross-system customer identity records require deduplication.
Timeliness	53%	Healthy	Batch ETL sync intervals (daily/weekly) create lag; real-time event streaming is required for autonomous agent response.
Consistency	51%	Needs Attention	Discrepant schema naming conventions across sales CRM vs ERP databases require canonical data modeling.
Accessibility	55%	Critical Blocker	Data locked in siloed department repositories without unified semantic vector embeddings or secure REST/GraphQL endpoints.

Data Preparation Budget Allocation

Data cleansing, OCR ingestion, vector embedding extraction, and metadata tagging represent **35% (₹22 - ₹32 Lakhs)** of the Phase 1 transformation budget. Allocating these funds early prevents expensive hallucination remediation during agent rollout.

2. Target Architecture Reference Pattern (Hypothesis)

Based on initial discovery findings across disparate document repositories and ERP ledgers, Nisol AI presents the following **Target Architecture Hypothesis**. *Note: Specific vector topologies (pgvector vs. Qdrant vs. native Snowflake/Databricks vectors) and CDC streaming mechanisms are subject to formal validation during the Phase 1 Technical Discovery Spike based on Novatech Systems Ltd.'s cloud VPC security boundaries and data residency mandates.*

Commercial Term: Client Infrastructure Pass-Through

All vector database hosting, cloud compute instances (AWS/Azure/GCP), and foundation model token consumption are direct pass-through expenses provisioned within Novatech Systems Ltd.'s enterprise cloud tenant. Nisol AI fees reflect custom pipeline engineering, prompt orchestration, and evaluation harnesses exclusively.

Vector Indexing & Embedding Model

Dense vector embeddings (BGE-M3 / OpenAI text-embedding-3-large) combined with sparse BM25 keyword indices inside pgvector / Qdrant with HNSW cosine similarity search.

Change Data Capture (CDC) Pipeline

Real-time Debezium / Kafka streaming connectors replicating modified CRM, ERP, and ticket tables into vector stores with automated chunk invalidation.

Recommended Data Engineering Roadmap

- Build a centralized Hybrid Vector Lakehouse (e.g. pgvector / Qdrant with hybrid BM25 search)
- Deploy an automated Document Ingestion & OCR pipeline with chunk-level metadata tagging
- Implement real-time Change Data Capture (CDC) pipelines from ERP/CRM into analytical vector stores
- Configure granular Role-Based Access Control (RBAC) filtering at the embedding chunk level

PILOT DECISION GATE & ACCEPTANCE PROTOCOL

PoC EVALUATION FRAMEWORK
& SCALABILITY DOSSIER

Pre-Execution Governance Protocol: Hypotheses, SLA Benchmarks & Production Gate Criteria

CONFIDENTIAL

ORGANIZATION

Novatech Systems Ltd.

STANDARD PILOT SPRINT

4-to-6 Week Active Engineering

Commences upon client dependencies fulfillment

GOVERNANCE GATE

GATE 0: PILOT SCOPE
AUTHORIZATION

DATE OF ISSUE

August 31, 2026

1. Pilot Hypotheses, Data Dependencies & Client-Side Checklist

Prior to authorizing live production deployment for any enterprise AI capability, Nisol AI mandates a controlled **4-to-6 week active sandbox pilot sprint**. This document functions as the **pre-execution governance protocol and rules of engagement**—establishing quantitative acceptance thresholds, client-side data prerequisites, stop-the-clock terms, and post-pilot production decision gates.

TARGET CITATION PRECISION

> 95.0%

Zero tolerance for uncited hallucinations

TARGET INFERENCE LATENCY

< 2.0s

Interactive UI SLA at p95

TARGET USER TASK REDUCTION

> 40.0%

Measured vs. human baseline

1.1 Candidate Pilot Workstreams & Required Data Dependencies

Candidate Workstream	Hypothesis to Validate	Required Acceptance Threshold	Required Data Source Dependency
Automated QA Test Generation Engineering & DevOps	Agent auto-generates executable Playwright tests from PR diffs	≥ 85% valid syntactical execution; zero broken existing tests	GitHub PR Webhook Access Read-only repo diff tokens + Playwright runner container
Intelligent Invoice OCR Parser Finance & Accounting	Vision LLM extracts 15 line items and matches ERP PO numbers	≥ 95% field extraction accuracy across 100 historical invoices	Historical Invoices & PO Ledger 100 historical vendor PDFs + ERP PO CSV export table
Enterprise SOP Knowledge Hub HR & Operations	RAG engine answers policy queries with exact PDF handbook citations	≥ 98% factual precision across 200 synthetic challenge queries	Verified Policy Repositories Unrestricted HR/Operations handbook PDFs + 200 SME QA pairs

1.2 Client-Side Dependency & Data Readiness Checklist (Prerequisites)

To guarantee sprint velocity and protect delivery margins, the **4-to-6 week Pilot Sprint clock shall commence only upon Nisol AI's formal written confirmation** that all prerequisite client-side dependencies have been satisfied:

Dependency Category	Required Deliverable	Responsible Party	SLA Deadline
Invoice Data Sample	100 historical vendor invoices (PDF/JPEG) + corresponding ERP PO numbers in CSV	Finance Controller	5 Business Days
RAG Golden Test Set	200 question-answer verification pairs reviewed by nominated internal SMEs	Ops / Legal SMEs	7 Business Days
API & Code Diff Access	Read-only tokens for GitHub PR diffs (Engineering) + Jira read access (PMO)	CTO / IT Ops Lead	10 Business Days
SME Time Allocation	Confirmed calendar holds for nominated SMEs (2 hours/week for testing & validation)	Department Heads	At Signing
Cloud Sandbox VPC	Client-provisioned VPC / Sandbox environment with foundation model API quotas approved	CTO / Cloud Ops	10 Business Days

Institutional Safeguard: Stop-the-Clock Clause

If client-side data delivery, credential provisioning, or SME availability slips beyond the agreed deadlines, the active pilot sprint clock automatically pauses until all prerequisites are validated in writing. This ensures delivery pod capacity is not stranded while awaiting internal client sign-offs.

2. Post-Pilot Production Go-Live Gates & Authorization Sign-off

At the conclusion of the 4-to-6 week active sandbox sprint, the joint Steering Committee evaluates empirical metrics against four strict **Post-Pilot Production Go-Live Gates** before authorizing live enterprise-wide deployment:

Decision Gate	Evaluation Method	Pass Condition	Accountable Lead
Gate 1: Accuracy & Fidelity	Automated evaluation against 200 ground-truth golden test cases	$\geq 95.0\%$ Accuracy	AI Solutions Architect
Gate 2: Security & PII Redaction	Adversarial red-teaming and prompt-injection stress testing	Zero PII Leaks	CISO / InfoSec Sponsor
Gate 3: End-User Usability	Department super-user evaluation across 5 business workflows	$\geq 80\%$ User Satisfaction	Department Business Sponsor
Gate 4: Unit Economics	Inference token cost per completed workflow calculation	Within Target OPEX Budget	Corporate Controller

Pilot Scope Freeze Protocol

The Pilot Sprint evaluates strictly the Candidate Workstreams enumerated in this dossier. Any additional capability requests, feature enhancements, or adjacent departmental use cases proposed during the sprint are formally logged for Wave 2+ scoping and shall not alter current sprint deliverables without a mutually executed Change Order.

No-Blame Early Termination (Client Right)

If at any point during the Pilot Sprint Novatech Systems Ltd. determines in its sole discretion that the AI capability is unlikely to achieve target acceptance thresholds, Novatech Systems Ltd. may terminate the pilot upon 5 business days' written notice. Nisol AI shall invoice solely for services actually rendered to date, and Novatech Systems Ltd. retains all generated prompts, test harnesses, and scripts at zero additional charge.

Formal Authorization Protocol: Gate 0 Approval

By executing below, Novatech Systems Ltd. authorizes Nisol AI to initiate the Sandbox Pilot Sprint under the terms, data dependency requirements, scope freeze boundaries, and quantitative acceptance criteria outlined in this dossier.

For: Novatech Systems Ltd. Evaluation Sponsor

For: Nisol AI Advisory

Lead AI Solutions Architect Signature & Date

Product Leader / VP Sponsor Signature & Date

CONFIDENTIAL